87

Midterm Examination of Electromagnetics, Fall 2025

(2025/10/29) Class: 電機 三 Student Id. Number: 411287037 Name: 劉東逸

Note: 不可使用任何電子裝置。**(Do not use any electronic equipment.)**

* The notations used in this test sheet follow those used in the textbook. ((x, y, z), (r, ϕ, z), (R, θ, ϕ): position in Cartesian, cylindrical, and spherical coordinates, respectively. ϵ: permittivity, ρ: charge density. $\mathbf{E}, \mathbf{D}$: electric field intensity, electric flux density. E_{1n}, E_{1t}: the normal component and tangential component of $\mathbf{E}$ in medium 1. $\mathbf{a}_n$: the unit normal vector pointing from the boundary of medium 2 to medium 1 when regarding boundary conditions.) Please also follow this convention.

A.[60%] Selection(選擇題): Please choose the most appropriate answer to fill in the brackets in front of each question. Answers in non-designated areas will not be scored. Option "none" means "none of the above". (請選出最適合的答案填入各題前面括號，作答於非指定區域者不予計分。選項 "none" 意指 "none of the above")

+60

1. (1) A lucite sheet ($\epsilon_r = 3.2$) is introduced perpendicularly in a uniform electric field $\mathbf{E}_o = \mathbf{a}_x E_o$ in free space. Determine the electric field intensity $\mathbf{E}_i$ inside the lucite. [1.]$\mathbf{a}_x(\frac{E_o}{3.2})$; [2.]$\mathbf{a}_x(3.2\epsilon_0 E_o)$; [3.] $\mathbf{a}_x(1 - \frac{1}{3.2})E_o$; [4.]$\mathbf{a}_x(1 - \frac{1}{3.2})\epsilon_0 E_o$; [5.] none.

$\varepsilon_0 \vec{E}_0 = \varepsilon_0 \varepsilon_r \vec{E}_i$

2. (3) Which of the following about the boundary conditions of the electrostatic field at the interface between two different dielectrics is wrong? [1.] $D_{1n} - D_{2n} = \rho_s$; [2.] $E_{1t} = E_{2t}$; [3.] $D_{1t} = D_{2t}$ under any circumstance; [4.] $D_{1n} = D_{2n}$, if there is no free charge available in both media; [5.] none.

+10

3. (3)In the cylindrical coordinate system, what is the result of $\frac{\partial \mathbf{a}_\phi}{\partial \phi}$? [1.]$\mathbf{a}_\phi$; [2.]$\mathbf{0}$; [3.]$-\mathbf{a}_r$; [4.]$\mathbf{a}_r$; [5.] none. $-\sin\varphi\,\hat{a}_x + \cos\varphi\,\hat{a}_y$ $-(\cos\varphi\,\hat{a}_x + \sin\varphi\,\hat{a}_y)$

4. (4) Consider a vector function $\mathbf{F} = \mathbf{a}_x(x + c_1 z) + \mathbf{a}_y(c_2 x - 3z) + \mathbf{a}_z(x + c_3 y + c_4 z)$. If $\mathbf{F}$ is solenoidal, then $c_4 =$ [1.] 1; [2.] 0; [3.]-3; [4.]-1; [5.] none. $1 + c_4 = 0$

+10

5. (4) Given the vectors $\mathbf{A} = \mathbf{a}_x + 2\mathbf{a}_y - 3\mathbf{a}_z$, $\mathbf{B} = -4\mathbf{a}_y + \mathbf{a}_z$, and $\mathbf{C} = 5\mathbf{a}_x - 2\mathbf{a}_z$, find the magnitude $|\mathbf{A} - \mathbf{B}|$ = [1.] $\sqrt{50}$ [2.] $\sqrt{21}$ [3.] $\sqrt{35}$ [4.] $\sqrt{53}$ [5.] none. $|\hat{a}_x + 6\hat{a}_y - 4\hat{a}_z| = \sqrt{36+1}$ +16

6. (5) In the cylindrical coordinate system, $\int_0^\pi \mathbf{a}_r d\phi$ = [1.] $\mathbf{0}$; [2.] $2\mathbf{a}_y$; [3.] $\pi\mathbf{a}_r$; [4.] $-2\mathbf{a}_y$; [5.] none. $\int_0^\pi (\cos\varphi\,\vec{a}_x + \sin\varphi\,\vec{a}_y)\,d\varphi$ $= [\sin\varphi\,\vec{a}_x - \cos\varphi\,\vec{a}_y]_0^\pi$ $\hat{a}_y + \hat{a}_y$

7. (1) Consider a system consists of three charges Q_1 at position $(x, y) = (-1, 0)$, Q_2 at position $(x, y) = (1, 0)$, and Q_3 at position $(x, y) = (0, 2)$. Assuming $Q_1 = Q_2 = q$, please calculate the charge of Q_3, so that the test point $P : (x, y) = (0, 1)$ is of zero electric field intensity, $\mathbf{E} = \mathbf{0}$. [1.] $\frac{q}{\sqrt{2}}$ [2.] $\sqrt{2}q$ [3.] $\frac{q}{\sqrt{3}}$ [4.] q [5.] none.

8. (1) Given a vector field $\mathbf{F} = \mathbf{a}_r k_1/r + \mathbf{a}_z k_2 z$ in cylindrical coordinate system, evaluate $\oint_S \mathbf{F} \cdot d\mathbf{s}$ over the surface of the closed cylinder defined by $z = \pm 3$ and $r = 2$. [1.] $12\pi(k_1 + 2k_2)$; [2.] $12\pi(k_1 + k_2)$; [3.] $12(k_1 + 2k_2)$; [4.] 0; [5.] none.

9. (2) Consider a vector function $\mathbf{F} = \mathbf{a}_x(x + c_1 z) + \mathbf{a}_y(c_2 x - 3z) + \mathbf{a}_z(x + c_3 y + c_4 z)$. If $\mathbf{F}$ is irrotational, then $c_2 =$ [1.] 1; [2.] 0; [3.]-3; [4.]-1; [5.] none.

10. (4) Given a scalar field $A(R, \theta, \phi) = R^2 \cos\theta + R\tan\phi$, at position $(R, \theta, \phi) = (2, \frac{\pi}{4}, \frac{-\pi}{4})$, $\nabla \times (\nabla A) =$ [1.] $\mathbf{a}_R 2\sqrt{2}$; [2.] $-\mathbf{a}_\theta\sqrt{2}$; [3.] $\mathbf{a}_R 2\sqrt{2} - \mathbf{a}_\theta\sqrt{2}$; [4.] 0; [5.] none.

11. (3) A straight line charge with uniform density ρ_ℓ is placed along the z-axis in the region $-\infty < z < \infty$. At a point $P : (r, \phi, z) = (r, \phi, 0)$, please find the electric field intensity $\mathbf{E}$. [1.] $\mathbf{0}$; [2.] $\mathbf{a}_r \frac{\rho_\ell}{6\pi\epsilon_0 r}$; [3.] $\mathbf{a}_r \frac{\rho_\ell}{2\pi\epsilon_0 r}$; [4.] $\mathbf{a}_r \frac{\rho_\ell}{4\pi\epsilon_0 r}$; [5.] none.

12. (2) A cloud of charges with density $\rho = \rho_0$ for $R \le 5$ and $\rho = 0$ for $R > 5$ is spherically distributed in air. If the charge cloud is centered at the origin of the coordinate system, what is the $\mathbf{E}$ value at position P: $(R, \theta, \phi) = (3, \frac{\pi}{4}, 0)$? [1.] $\mathbf{a}_R \frac{\pi\rho_0}{12\epsilon_0}$; [2.]$\mathbf{a}_R \frac{\rho_0}{\epsilon_0}$; [3.]$\mathbf{a}_R \frac{125\rho_0}{27\epsilon_0}$; [4.]$-\mathbf{a}_R \frac{125\rho_0}{27\epsilon_0}$; [5.] none.

B.[40%] Calculations (計算題): Please answer the following questions in the blank area of this test sheet. Remember to mark the question numbers. Write the calculation process clearly. No points will be given for answers only. (下列題目請標明題號作答於本試題紙空白處，清楚寫出計算過程，只寫答案不予計分。)

+10

13. (10%) Determine the boundary conditions for the tangential and the normal components of the polarization vector $\mathbf{P}$ at an interface between two perfect dielectric media with dielectric constants ϵ_{r1} and ϵ_{r2}.

+10

14. (10%) A line charge of uniform density ρ_ℓ in free space forms a semicircle of radius b. At the center of the semicircle, determine (a) the electric potential V, and (b) the electric field intensity $\mathbf{E}$.

+2

15. (10%) In free space, an infinitely long cylindrical shell of charge is centered on the z-axis and extends between $r = a$ and $r = b$. The volume charge density within this shell is uniform and equal to ρ_v. Find the electric field intensity $\mathbf{E}$ everywhere (Note: There are three regions to consider)

+9

16. (10%) Three charges ($+q, -2q$, and $+q$) are arranged along the z-axis at $z = \frac{d}{2}$, $z = 0$, and $z = -\frac{d}{2}$, respectively. At a distant point $P : (R, \theta, \phi)$, please calculate (a) the electric potential V, and (b) the electric field intensity $\mathbf{E}$.

13

$\vec{P} = \vec{D} - \varepsilon\vec{E}$

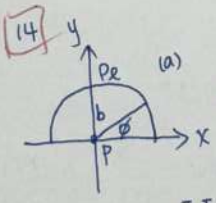
$D_{1n} = D_{2n}$

$E_{1t} = E_{2t}$.

$\varepsilon_{r1} \mid \varepsilon_{r2}$.

0

14



(b)

$dV = \frac{dQ}{4\pi\varepsilon_0 R}$

$d\vec{E} = \frac{dQ\,\vec{R}}{4\pi\varepsilon_0 |\vec{R}|^3}$

$V = \int_0^\pi \frac{\rho_\ell b\,d\phi}{4\pi\varepsilon_0 b}$

$= \frac{\rho_\ell}{4\varepsilon_0}$ (V) #

因為P點在圓心，半圓要對P點的水平電場互相消 ⇒ 只需算y方向

10

$\vec{E} = \int_0^\pi \left[\frac{\rho_\ell b\,d\phi}{4\pi\varepsilon_0 b^3} b\sin\phi\right][\hat{a}_y]$

$= \frac{\rho_\ell}{4\pi\varepsilon_0 b}\int_0^\pi \sin\phi\,d\phi\,[-\hat{a}$

$= \frac{-2\rho_\ell}{4\pi\varepsilon_0 b}\hat{a}_y$

$= \frac{-\rho_\ell}{2\pi\varepsilon_0 b}\hat{a}_y$ (V/m) #